

BABY BOOMLETS AND BABY HEALTH: HOSPITAL CROWDEDNESS, TREATMENT INTENSITY, AND INFANT HEALTH*

Mindy Marks¹ Moonkyung Kate Choi

Abstract

A challenging question in health economics concerns the benefit of additional treatment. We propose a new approach for estimating the marginal returns to medical spending based on the non-uniform distribution of birth dates. The thought experiment is that on a relatively crowded day an infant may receive reduced care because resource constraints are binding. Specifically, we compare infant health within the same hospital using variation in hospital charges per birth that arises from short-term hospital crowding after netting out time period (day of the week, month of the year, and calendar year) effects. Using detailed information on every hospital birth in California from 2002 to 2006, we first present evidence that the level of hospital crowdedness is orthogonal to underlying infant characteristics that may have independent impacts on spending and health. We then document that treatment is less intensive when the birth hospital is more crowded. Using hospital crowdedness as an instrument for health care spending, we show that OLS estimates overestimate the benefits of medical care. Our results suggest that on the margin the health benefits from additional spending are negligible. In particular, additional spending does not reduce either neonatal (28 days) or one-year infant mortality rates. These findings are robust to alternative measures of hospital crowdedness and treatment intensity and persist for the subsample of low-birth-weight infants. *JEL* codes: I12, I18

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1. Department of Economics
University of California, Riverside
4110 Sproul Hall, Riverside, CA 92521, USA
Fax: (951) 827-5685
Tel: (951) 827-4164
E-mail: mindy.marks@ucr.edu

I. Introduction

Health care spending consumes an ever larger portion of US GDP.¹ Existing evidence about the benefit of the marginal dollar of medical spending, both overall and for newborns in particular, is mixed. Evidence from studies investigating the effectiveness of new therapeutic improvements tends to suggest that additional spending generates declines in overall and infant mortality. (Williams and Chen, 1982; Cutler, 2005; Phibbs et. al., 2007; Richardson et. al., 2007; Almond et. al., 2010) However other studies suggest that the new technologies have expanded to the point where additional benefits are negligible. (Goodman et. al., 2002; Almond and Doyle, 2011)

A methodological challenge when identifying the causal relationship between health care spending and health is non-random selection of patients into treatment. To clearly identify the relationship between treatment intensity and infant health, an exogenous source of variation in treatment intensity is needed. To generate exogenous variation in health care spending, this paper introduces a new identifying variable: hospital crowdedness measured by the number of infants born on a given day in a given hospital. We examine the effectiveness of additional treatment that stems from the non-uniform distribution of birth dates within a given hospital.² The thought experiment is that on a “slow” day an infant may receive more care either because the resource constraints are less binding or because health care providers respond to the temporary income shock by performing additional procedures.

¹ In 2010, 17.6 percent of GDP was spent on health care. Childbirth is the most common medical procedure. According to Nationwide Inpatient Sample data (HCUP, 2005), almost 19 percent of all hospitalization were related to childbirth in 2005. Also, the average cost for newborn delivery is high and has been rising. (Russell et al., 2007) During the sample years, the average hospital charge for delivery went up from \$10,989 in 2002 to \$13,603 in 2006. Moreover, the second and third most expensive condition treated in US hospitals were “Mother’s pregnancy and delivery” and “newborn infants, which accounted for 5.2 percent and 4.3 percent of the national hospital bill in 2004, respectively. (Russo and Andrews, 2006)

² This approach is similar in spirit to Hoxby (2000) who looks at the effect of class size on student achievement using exogenous variation in class size that stems from idiosyncratic variation in the population.

We compare infant health within the same hospital using variation in hospital charges per birth that arises from short-term hospital crowding after netting out time period (day of the week, month of the year, and calendar year) effects. Variation in hospital crowdedness that stems from heterogeneity in capacity across hospitals or within hospital variation in crowdedness attributable to the preference towards weekday births would induce an artificial correlation between crowdedness and infant health. By exploiting the variation in crowdedness within the hospital and time period, our estimates of the effects of treatment on infant health are free of potential bias from the variations in resource availability, resource quality, or patient mix among hospitals and across time periods.

Previous studies have used legislation changes to obtain estimates of the causal impact of additional treatment on birth outcomes. Unlike these studies our empirical approach exploits variation in treatments that stems from decisions made by medical personnel when responding to hospital capacity constraints. In other words, we are identifying the impact of care medical personnel chose to add when unconstrained, which is likely to be of more value than care added by legislation. Existing studies utilizing mandates on the mandatory minimum length of hospital stay measure the marginal benefit of an extra day for those infants who would have otherwise had been discharged by medical personnel and find little impact.³ In other words, if the health care providers had been practicing effectively, it is not surprising that the extended hospital stay

³ Using legislation mandating coverage of minimum postnatal hospital stays, Meara et. al. (2004) find rates of rehospitalization for jaundice within 10 days fell in the year after legislation was introduced, but rates of all-cause rehospitalization, dehydration, and infection diagnoses did not change. Madden et. al. (2002) find no health effect of extended hospital stay among infants with normal vaginal deliveries. Evans et. al. (2008) also find that additional one day hospital stay decreases readmission rate within 28 days by about one percentage point for subsamples of vaginal delivery with complications and cesarean delivery without complications. But they did not find any impact of mortality rate in any of the subsamples. Almond and Doyle (2011) find that infants born 10 minutes after midnight do not show better health status than infants born 10 minutes before midnight, despite longer hospital stay due to hospital billing practice. Moreover, using California's minimum-insurance mandate which required insurance coverage for at least two days of hospitalization after birth, they find that no health gain is realized in either the 1 to 2 nights stay margin or the 2 to 3 nights stay margin.

for a healthy infant (who would have been voluntarily discharged before the mandate) does not produce any health gains. Using state-level variation in tort reform, Currie and Macleod (2008) find no impact of a reduction in procedures such as caesarean section and inducement on newborn health. Again, a null of effect of treatment intensity on newborn health outcomes might be expected because the procedures identified are the procedures eliminated because of the lessened fear of malpractice lawsuits (i.e. procedure of marginal medical value which the medical community was only performing out of fear of lawsuits).

Additionally, we focus on the causal relationship between health care charges and infant mortality.⁴ Hospital charges contain comprehensive information on the hospital care infants receive. As the best available summary measure of health inputs (Almond et al., 2010), hospital charge reflects length of stay, number of procedures, and kinds of procedures performed during the hospitalization. Health care spending is also at the center of the policy debates. In addition we estimate a policy relevant Local Average Treatment Effect. The infants who receive less care when the hospital is crowded are precisely the infants with marginal health needs who would be the first ones to receive less care if hospitals became more resource constrained due to hospital closures or population growth.

This number of infants born on a given day in a given hospital is a plausible instrument because the number of other infants who share the target infant's birthday should not have any independent impact on the target infant's health other than through the intensity of hospital care reflected in hospital charges. Moreover, the number of other infants born on a given day in a

⁴ Throughout the paper we will use the terms charges and spending interchangeably. Hospital charges include all charges for services rendered during the infants stay at the facility, based on the hospital fully established rates. Hospital-based physician fees are excluded. We assume that hospital charges and physician fees are positively correlated. Additionally hospital charges could be thought of as list prices, the actual prices are often much lower and vary by hospital and insurance companies. However we have no reason to believe that the negotiated price varies by the daily hospital crowdedness. Also, we control for hospital fixed effects and type of insurance coverage in all regressions.

given hospital is highly correlated with health care spending for the target infant. Our study suggests that for each standard deviation increase in the number of births in a given hospital-day of the week, the average hospital charge for infants born on the same day decreases by \$65 and by \$1,463 for the subsample of low-birth-weight infants⁵. Moreover, summary statistics show that after conditioning on the day of the week, infants born on slow versus busy days have very similar levels of prenatal care, parents' demographic characteristics, proxies for infant health, labor complications, and insurance status. Hospital crowdedness thus mimics an experiment in which nature assigns, to each infant, a level of treatment intensity, independent of her background.

Our dataset captures over two million births that occurred in California hospitals between 2002 and 2006. OLS estimates suggest that higher health care spending is associated with better infant health. Our main finding is that infants who had more intensive hospital stays because they were born on slow days fared no better in health status than their busy day counterparts. Additional health care spending does not improve infant health status as measured by neonatal (28 days) mortality or one-year mortality rates. These findings are robust to alternative measures of crowdedness and for the subsample of low-birth-weight infants. When we examine the impact of length of hospital stay or include spending on the birth mother, both of which also responds to hospital crowdedness, similar null findings are obtained. Hospital crowdedness has a larger impact on treatment intensity in relatively under-resourced hospitals, yet even in the most resource constrained hospitals, the IV estimates show that reductions in treatment do not negatively impact infant health. Finally, we provide evidence in support of binding capacity

⁵ The change in treatment intensity identified by our empirical approach is small relative to papers that exploit changes in treatment that stem from insurance coverage. These papers, which do not cover infant mortality since the vast majority of births are covered by insurance, suggest that additional treatments improve mortality. (Doyle, 2005; Card et al, 2009)

constraints as opposed to supplier induced demand for the change in treatment associated with hospital crowdedness.

The closest paper to this study is Freedman (2012). Freedman’s paper documents that the probability of admissions to a Neonatal Intensive Care Unit (NICU) increases when there are more vacant NICU beds in an infant’s delivery hospital the day prior to birth. The effect is particularly large for low-birth-weight infants for whom there is discretion over the appropriate amount of treatment. These results are in alignment with our findings that hospitals change their treatment decisions in response to short-term variations in capacity constraints. While interested in the causal impact of NICU admission on mortality, the author argues that it is inappropriate to use an instrumental variables approach to estimate the effect of NICU admissions on hospital charges and health outcomes with empty NICU beds as an instrument. First, NICU capacity may impact costs and outcomes through avenues other than NICU admission, say by substitution of resources between NICU and non-NICU patients. Since we focus on the impact of changes in hospital spending and not changes in NICU admissions in response to crowdedness, our estimates do not suffer from the above problem. Second, infants who are placed into a NICU because there are empty beds are likely to be unobservably healthier and thus have lower mortality rates.⁶

This paper is organized as follows: Section II describes the dataset and presents the identification strategy. Section III discusses the relationship between hospital crowdedness and

⁶ Reduced form estimates in Freedman (2012) suggest that infants born on days with vacant NICU beds have lower mortality rates, thus suggesting that additional capacity in the form of NICU beds would increase infant health. If we estimate a reduced form equation similar to the one estimated by Freedman, we find that infants born on slow days have higher (but not statistically significant) mortality rates. If infants who receive extra care when the hospital has vacancies are unobservably healthier (as is likely to be the case when care is defined as NICU admission and could be the case when care is measured by additional spending) then estimates of treatment on infant mortality are likely to be downward biased. Since we find a negative relationship between treatment and infant health, we can rule out a positive treatment effect.

receipt of care. Section IV presents our main findings. Section V presents and discusses results from various robustness checks. Section VI presents evidence that the variation in treatment intensity is due to capacity constraints not to supplier induced demand. Section VII concludes the paper.

II. Empirical Strategy

A. Data

The data used in this study are confidential data provided by the California Office of Statewide Health Planning and Development (OSHPD). The OSHPD data link infant hospital discharge records for all hospital stays during the first year of life with infant vital statistics data (birth and death certificate data). The OSHPD data used in this paper include every hospital birth in California between 2002 and 2006. The OSHPD data provide birth date and birth hospital, which are used to generate our identifying variable, the hospital-level crowdedness on delivery day. While lacking complete health procedure information, the OSHPD data contains hospital charges for both infant and mother and length of hospital stay. The OSHPD data also provide detailed information on prenatal care, parents' demographic information, and newborn characteristics. The OSHPD data also contain our outcome variables – neonatal and one-year mortality.

For the purposes of this study, we measure hospital charges/spending as the sum of hospital charges from all consecutive hospital stays after birth. If a newborn was transferred from the birth hospital, we track the infant charges for all transferred hospitals stays until the newborn was discharged and included these charges in our measure of infant hospital charges. Similarly, hospital stay is the length of consecutive hospital stays after the birth including transfers.

Initial data from OSHPD contained 2,675,440 birth records with birth date and birth hospital. We exclude about 0.6 percent of observations for which hospital discharge data are not linked to vital statistics data.⁷ About 11.6 percent of the birth records were missing hospital charge information. About 99 percent of births with missing charge information are Kaiser births because all Kaiser Foundation Hospitals are exempt from reporting hospital charges. Instead of charging specifically for an inpatient stay, Kaiser Hospitals receive a constant monthly (capitated) payment from each member, whether or not that member is hospitalized. Thus we exclude the Kaiser Hospitals from the main analysis. Additionally, a small fraction of our sample (less than three percent) is born in a hospital where the average number of births per day is less than two. Because there is limited variation in the number of births, we exclude births from these hospitals. We also drop infants born during the first and the last weeks of our sample period. When we extend our measure of crowdedness, we cannot obtain the number of infants born before or after the delivery day for those infants born on the first or last week of the sample period, respectively.⁸ Our final sample contains 2,287,119 infants that are born in one of 226 California hospitals between the years 2002 and 2006.

Table I reports the summery statistics for our sample as well as the subsample of infants with birth weight less than 2500g. The average infant in our sample is born to a 28 year-old women who completed high school. Over half of the births in our sample are to Hispanic women and around half of the births are covered under Medicaid. Average infant hospital charges are around \$13,000 per birth. Hospital charges for low-birth-weight infants are 8.8 times higher than charges for all infants. Charges increase to \$27,041 if we include hospital charges billed to the

⁷ These missing observations appear balanced across busy and slow days. Additionally we retain the observations for infants who died at the birth hospital since we know the mortality outcome.

⁸ Before dropping any of the observations, we include them to calculate the hospital crowdedness measure.

mother for the delivery and follow-up care. The average length of stay in hospital is 3.36 days which rises to 16.5 days for low-birth-weight infants. While one-year mortality of all infants is 0.5 percent, that of low-birth-weight infants is 4.7 percent. While only 3 percent of all births are multiple births, about 25 percent of low-birth-weight births are multiple births.

Since we generate the variation in hospital spending from the different levels of hospital crowdedness, our identification strategy requires sufficient within hospital variation in the daily number of births. Panel A in Table II reports descriptive statistics of number of births for selected representative hospitals. Based on the average daily number of births, the smallest, every 10th percentile, and the largest hospitals are selected. Column (1) shows the mean average daily number of births. Column (2) contains the standard deviation. The third through seventh columns report the minimum, the quartiles, and the maximum number of infants born a day for each representative hospital during our sample period. Table II shows there is sizable variation in the number of births within a given hospital. For example, in the 50th percentile hospital, 20 infants were born on the busiest day and only one infant was born on the slowest day. The interquartile ranges are also sizable.

While these numbers give a sense of the underlying variation in crowdedness, the identification strategy is based on variation after netting out the hospital and day of the week specific effects. We condition on days of the week, because there are significantly more births on weekdays than weekends. Numerous authors have documented an association between weekend births and higher infant mortality rates (MacFarlane, 1978; Rindfuss et. al., 1979; Hendry, 1981; Mangold, 1981; and Mathers, 1983). Failure to condition on the day of the week would thus induce a correlation between crowdedness and health. The first row of panel B in Table II summarizes the variation in the residuals from a regression of the number of birth on the

complete set of hospital and day of the week fixed effects. At the 25th percentile there are 1.97 fewer daily births than the hospital-day of the week average, at the 75th percentile there are 1.84 more births than the hospital-day of the week average. Throughout the paper we refer to *slow* days as days with residuals in the bottom quartile and *busy* days as days with residuals in the top quartile. The standard deviation of the residual is 3.13 (by construction the mean is zero). Throughout the paper we will discuss the magnitude of the first stage results in the context of change in the number of births by one standard deviation of the residual (residual crowdedness).

B. Validity of the Instrument

The underlying assumption for our identification strategy is that hospital crowdedness should be uncorrelated with any infant level observable (and unobservable) trait that may impact infant health. For instance, say that extremely warm weather creates pregnancy complications that induce early labor. If so warm days would be more crowded, and those infants born on warm days would need more treatments due to shorter gestation lengths, thus threatening our identification strategy. Another possible threat would be hospitals rerouting mothers in labor. If crowded hospitals direct mothers to less crowded hospitals based on potential delivery complications, our estimates might be biased because the underlying health status of infants born would be correlated with hospital crowdedness.

Table III presents evidence on the validity of our instrument. The table reports summary statistics for infants born on *busy* days and *slow* days in each hospital-day of the week. In essence, Table III asks, conditional on day of week and hospital, whether infants born on *slow* days are different for their counterparts born on *busy* days in terms of observable characteristics. If there are differences in observable traits, it calls into question the identify strategy as it

suggests that *slow* day birth may differ in their health outcomes for reasons other than hospital crowdedness. The average *busy* day has about twice as many births as the average *slow* day. While same day transfers to a different hospital are slightly more likely for *busy* day infant, the likelihood of transfer for low-birth-weight infants is independent of hospital crowdedness. Thus the data suggest that hospitals are not rerouting the sickest infants elsewhere on crowded days.

The rest of the table demonstrates that for many important infant background variables *slow* day births mirror their *busy* day counterparts. *Slow* day infants and *busy* day infants received similar levels of prenatal care. Parents of *slow* day infants tend to be younger and less educated. However the size of the difference is very small. For example, mothers of *slow* day infants are younger than mothers of *busy* day infants by 0.03 year, which is less than half a month difference. The difference in maternal education is only 0.05 years. Differences in paternal education and age are even smaller. The gender and racial distribution of infant born on *slow* and *busy* days are almost identical. Twice as many multiple births happen on *busy* days, but this is to be expected because all else equals, multiple births lead to hospital crowdedness. Thus it is important to condition on multiple births in our analysis. Insurance status shows very similar coverage across *slow* and *busy* day births. Generally, the differences in observable traits between infants born on *slow* and *busy* days are negligible.

Given that we find that the reduction in spending associated with *busy* days does not harm infant health, we would be concerned if *busy* day infants are healthier than *slow* day infants. Birth weight is one of the most commonly used indicators of newborn health. Infants born on *slow* days are slightly heavier by 32 grams, implying that infants born on *busy* days are, if anything, slightly less healthy and more in need of medical attention (a result which persist in the subsample of singleton births). Gestation length is another indicator of infant health. *Slow* day

infants and *busy day* infants report identical length of gestation. The Caesarean section rate is slight higher on *busy day* birth, thus we control for Caesarean sections in all specifications.

Overall, the summary statistics show that, once we condition on the day of the week and hospitals, there are no apparent differences in observable family background and health indicators. This suggests that the level of crowdedness is orthogonal to underlying infant characteristics that may impact spending and health. Hospital crowdedness thus mimics an experiment in which nature assigns, to each infant, a level of treatment intensity, independent of her background.

C. Identification Strategy

To address the potential endogeneity of the health care spending, we use an instrumental variables strategy which utilizes the difference in health care spending that arises from the variation in the number of infants born on a given day in a given hospital. The first-stage equation for the IV estimate is:

$$\text{Spending}_{ijt} = \theta_1 \text{Crowded}_{ijt} + X'_{ijt} \theta_2 + \lambda_j + \varphi_t + \omega_{ijt} \quad (1)$$

where Spending_{ijt} is log of hospital charge for infant i born in hospital j on day t . If infant i was transferred to another California hospital after delivery, we tracked all transferred hospital stays and added all listed hospital charges to construct the infant spending measure. The key variable Crowded_{ijt} captures the number of infants born on infant i 's date of birth in hospital j on day t .

X'_{ijt} is a vector of control variables. Included in X'_{ijt} are pregnancy characteristics (the number of prenatal care visits, month the prenatal care began, and an indicator whether there were any pregnancy complications), parental characteristics (categorical dummy variables for age and education of mother and father, insurance status), and newborn characteristics (gender,

race, parity, and an indicator for whether the infant was the first born). Birth characteristics such as birth weight, length of gestation in days and whether caesarean section was performed are also included.⁹ Because of the nonlinear impact on infant health, birth weight is categorized at 500g intervals and length of gestation is categorized at two week intervals.

Also included in X'_{ijt} are indicator variables for the day of the week, month of the year, and four holiday indicators. As mentioned earlier, there is an established association between weekend births and higher infant mortality rates. If lower levels of staffing on weekends generate poor health outcomes, then not controlling for the days of the week would bias result because fewer infants are born on weekends and these infant have poor outcomes. However, more recent works (Dowding et. al., 1987; Gould et. al, 2003; Hamilton and Restrepo, 2003) show that difference in underlying infants' health and family background across weekend and weekday birth can account for the difference in mortality rates. To purge our result of these potential problems, we control for the day of the week. We also control for the month of the year the infant i was born. There is seasonal variation in crowdedness as more children are born in summer months than in other months of the year. Studies suggest that maternal characteristics are not uniformly distributed throughout the year. (Dehejia and Muney, 2004; Buckles and Hungerman, 2012). Currie and Schwandt (2013) show that there are significant differences in gestation and birth weight among infants conceived in different months even after controlling for maternal characteristics using mother fixed effects. We include indicator variable for each month

⁹ We recognize caesarean section surgery is potentially endogenous to the number of infants born on a given day in a given hospital, and may be especially responsive to SID (see Baicker et. al. 2006). To investigate the sensitivity of our results we have estimated regressions excluding caesarean section from the set of control variables. While the first stage results are larger, the results are very similar to our main findings. We have also directly investigated if crowdedness induces more caesarean sections. The regression result suggests that slightly fewer caesarean sections are performed on crowded days. Since most studies find no health benefit of caesarean section on infants, treatment of caesarean sections should not cause bias in our second stage analysis (Gruber and Owings, 1996). Finally, our results are robust to restricting our sample to vaginal delivery infants.

of the year to account for possible systematic seasonal differences in infant health. We also control for four major holidays (New Year's Day, Independence Day, Thanksgiving Day, or Christmas) when we observe a significant drop in the number of infants born. We suspect that the level and quality of hospital staffs and hospital care might be different on holidays.

λ_j and φ_i are hospital and year fixed effects. Hospital fixed effects control for any hospital specific characteristic that may impact infant health. This includes variation in level of technology, training of staff, resource capacity and patient mix. (Phibbs, 1996) The year fixed effects control for the possible differences in health care price, resource capacity, and medical knowledge over time which may be correlated with crowdedness. Robust standard errors are clustered at the hospital times year level.

III. Impacts of Crowdedness on Hospital Spending

Figure I.1 shows a visual representation of the first stage results. The dotted line displays the average infant health charges for infants born on *busy* days by 300 gram birth-weight bins. The solid line represents the analog for infants born on *slow* days. For ease of comparison, Figure I.2 reports the differences in health care spending between *busy* and *slow* day infants by birth weight bins. For the majority of births, it is clear that infants born on slower days receive additional spending when compared to their *busy* day counterparts.

While the first stages results are significant for the entire distribution of births and the subsample of births with normal birth weights, Figure I.2 suggests that the infants who receive more health care spending on *slow* days are disproportionately low-birth-weight infants (below 2500 grams). Moreover, the differences in hospital charges between low-birth-weight infants

born on *slow* and *busy* days are sizable. Given the sizable difference in spending, as a robustness check we conduct a separate analysis on the low-birth-weight subsample.

Table IV reports the first stage regression result from equation (1). The regression result suggests that when the number of infants born on a given day in a given hospital increases by one, hospital spending per birth decreases by 0.16 percent. Since the average infant hospital charge is \$12,985, a one standard deviation increase in the number of births in a given hospital-day of the week is associated with a decrease in average hospital charge of \$65. The first stage F-statistics is 19.6 suggesting our IV (hospital crowdedness) is highly correlated with hospital charge. We note that the reduction in spending does not simply reflect that the fact that infants born on relatively uncrowded days have longer hospitalization stays. If we include length of stay in equation (1) the coefficient on number of daily births decreases by 25% but remains statistically significant at the one-percent level. The first stage result provides strong evidence that health care providers change the intensity of treatment based on the short-term fluctuations in crowdedness in the hospital.¹⁰

IV. Estimating the Effectiveness of the Health Care Spending on Newborn Health

Figures II.1 and II.2 provide preliminary evidence that the additional spending that occurs on *slow* days does not translate into better health outcomes. Figure II.1 shows one-year mortality rates for infants born on the *busy* days and *slow* days for the same 300 gram bins used in Figures I.1 and I.2. For ease of comparison, Figure II.2 reports the differences in infant mortality rate between *busy* and *slow* day infants by birth weight. Despite the higher spending documented in

¹⁰ We explored the possibility of heterogeneity in the intensity of treatment for the subgroups of Medicaid and minority births (i.e. when the hospital is crowded is there a disproportionate decrease in spending for certain identifiable subgroups). We find no evidence of difference in treatment intensity in these subgroups in response to hospital crowdedness.

Figure I.2, *slow* day infants seem to, if anything, have higher mortality rates than their *busy* day counterparts. As is clear in Figure II, the additional spending that *slow* day infants receive does not appear to translate into lower mortality.

To formally measure the causal effect of health care spending on infant health, we use equation (2) for the second stage estimation in our two-stage least square (2SLS) model:

$$Y_{ijt} = \beta_1 \widehat{\text{Spending}}_{ijt} + X'_{ijt}\beta_2 + \mu_j + \tau_i + \varepsilon_{ijt} \quad (2)$$

where the dependent variable Y_{ijt} is an indicator of the health of infant i who was born in hospital j on day t . We employ neonatal and one-year mortality rates as measures of infant health.¹¹ $\widehat{\text{Spending}}_{ijt}$ is predicted log of hospital charge for infant i from equation (1). The rest of the control variables are the same as in equation (1). Given that our outcome variable is dichotomous, we estimate probit models and present marginal effects evaluated at the mean.

OLS estimates of β_1 are likely to be biased due to unobserved variables in ε_{ijt} . The direction of bias is given by two elements: the relationship between the omitted variable and the outcome variable (Infant Mortality), and the relationship between the omitted variable and the variable of interest (Spending). Consider the case where parental characteristics such as cautiousness/responsibility influence infant health. If parental cautiousness correlates negatively with infant mortality and positively with spending at delivery (say, because cautious parents tend to have better insurance and/or demand additional or more expensive treatments at birth for their newborns), excluding this variable from equation (2) biases the OLS estimates downwards since part of the estimated beneficial effect of spending on infant mortality can be attributed to parental cautiousness. On the other hand, less healthy infants may receive more intensive

¹¹ We tried restricting the outcome variable to mortality rate from non-accidental causes only (excluding ICD-10 code 295-350). The results are similar to those when mortality from all causes is outcome variable.

hospital treatment and have higher mortality rates biasing estimates of the returns to such care upward in situations where the medical practitioners can observe richer health controls than the econometrician. Since we have rich initial health measures, we expect the bias due to omitting parental characteristics such as awareness of medical procedures will dominate and that there will be downward bias in the relationship between treatment intensity and infant mortality. We will use hospital crowdedness to instrument for Spending_{ijt} in equation (2) to address the endogeneity concern.

Table V reports the estimated marginal effects of hospital spending (at the mean) on infant mortality from a probit model where we first ignore and then address the endogeneity of health care spending. Column (i) in Table V reports the OLS estimates of the effectiveness of health care spending for the two infant health outcomes. The reported marginal effects suggest that additional hospital spending leads to a small but statistically significant decrease in both neonatal and one-year mortality rates. Column (ii) contains the corresponding 2SLS estimates, which suggest that the effect of hospital spending on infant health, using the variation in spending that arises from the relaxation of a capacity constraint, is negligible. This is a key result of the paper. The additional spending on the infants who were born on slower days does not improve infant health as measured by neonatal or one-year mortality rates. Although statistically insignificant, the coefficients on the infant mortality measures are positive implying that, if anything, the additional health care spending harms the infants: a phenomenon known as iatrogenic harm. Numerous studies (Black, 1998; Ashton et. al., 2003; Fisher et. al., 2003; Jha et. al., 2003) find evidence consistent with iatrogenic harm.¹² These studies suggest that additional

¹² Using a major reform of the Department of Veterans Affairs health care system, Ashton et. al. (2003) and Jha et. al. (2003) find that the major reduction in hospital utilization does not cause any adverse health consequences. Utilizing wide regional variations in per capita Medicare spending and practice, Fisher et. al.

medical care might be harmful to patients because all treatments entail some risk. Additionally, greater use of diagnostic tests may find abnormalities which would not have caused harm and longer hospital stays increase the risk of infections.

Our IV estimates are identified off of infants who have additional charges billed solely because they were born on days with excess capacity. If hospitals triage care, then the infants who receive additional treatment when the hospital has excess capacity will be unobservably healthier and therefore should have lower mortality rates. Under triage, the estimates are biased towards finding a positive treatment effect. Given that the Table V reports **increased** infant mortality associated with additional spending, we conclude that, on the margin, additional spending does not reduce infant mortality. In the next section we investigate the robustness of this finding.

V. Robustness Tests

a. Capacity Heterogeneity

Within Californian hospitals there is significant heterogeneity in their underlying base capacity. If crowdedness decreases access to care, for hospital with relatively low capacity the impact of additional births on access to care should be greater than in hospitals with ample capacity. Additionally, lower capacity hospitals should be to the left of their high capacity counterparts on a health production function. As such while additional spending does not appear to translate into infant health gains at the average California hospital, it may improve health in the subsample of low-capacity hospitals. To investigate, we collected data from OSHPD's Hospital Annual Utilization on the number of neonatal infant care (NICU) beds in each hospital in each year. We

(2003) find that higher spending is associated with lower quality of care. Black (1998) warns the possibility of pseudo disease caused by overuse of diagnostic tests.

then constructed a measure of NICU beds per 1,000 births for each hospital-year in our sample and split the sample into those hospital-years above/below median level of NICU capacity. We treat NICU beds as a proxy for overall delivery capacity.¹³ As is shown in Panel C of Table VI, while the initial health of infants born in low and high capacity hospitals is similar, deliveries that occur in low-capacity hospitals have significantly lower charges (\$8,051 vs. \$17,939) and shorter lengths of stay (2.81 days vs. 3.91 days).

Panel A of Table VI shows that estimates of equation (1) when the sample is split into low and high capacity hospitals. As would be predicted if exceeding capacity results in reduced care, the impact of crowdedness on infant charges are concentrated in the low-capacity hospitals. The effect of an additional birth on spending is three times greater in low than high capacity hospitals. Additionally, the association between daily births and access to care is statistically insignificant in the subsample of high-capacity hospitals.

Panel B of Table VI contains the estimated marginal effects of health spending on infant mortality by hospital capacity. For the subsample of high-capacity hospitals, we find a statistically insignificant positive association between spending and infant mortality. Consistent with the notion of health production function, the marginal benefit of treatment is greater for the subsample of low-capacity hospitals, although the difference is statistically insignificant. Even in relatively low-capacity hospitals, additional treatment intensity does not appear to improve infant health measured by neonatal and one-year mortality rates.

¹³ One hundred eight hospitals have no NICU beds and are included in the low-capacity category. If we exclude these hospitals from the low-capacity category we obtain similar results. We also collect data on NICU by country and conduct the analysis for high and low-capacity counties. Consistent with our main analysis, we find that the effect of an additional birth on spending is over 2.5 times greater for births in hospitals in a low-capacity county. The association between hospital charges and infant mortality is insignificant in both high and low-capacity counties.

b. Alternative Measures of Crowdedness

Thus far, crowdedness is measured by the number of infants born on a given day in a given hospital. However, a newborn might also compete with infants born a few days before and/or after her birth date for medical resources such as hospital beds, access to medical procedures, and hospital staffs. Hence we create two alternative measures of crowdedness: the *expected number of infants on delivery day* and the *expected number of infants during stay*. Equation (3) and (4) show how we constructed these measures.

$$\text{Expected Number of Infants on Delivery Day}_{ij} = \sum_{s=0}^7 \omega_s \times \text{Number of infants born}_{ijs} \quad (3)$$

$$\text{Expected Number of Infants during Stay}_{ij} = \sum_{r=-7}^7 \omega_r \times \text{Number of infants born}_{ijr} \quad (4)$$

Expected Number of Infants on Delivery Day_{ij} is the weighted sum of infants born s days before infant i 's birthday in hospital j . ω_s is a weight. It is the average percentage of s day old infants in the hospital. For example, on the infant's birthday, s is 0 and the weight (ω_0) is 1. When s is 1, we determine the number of infants born the day prior to the target infant's birth date in hospital j and set ω_1 to 0.72, because according to our data, 72 percent of infants remain in the hospital one day after they are born. We account for the births that happened up to seven days before. By the seventh day, more than 96 percent of infants are discharged from hospitals and the weight of the number of infants born seven days before (ω_7) is 0.04. Expected Number of Infants during Stay_{ij} is constructed in a similar manner. It accounts for the number of infants born up to seven before and seven after infant i 's birth date. Thus infants born in the days after the target infant's birth will also be allowed to impact target infants access to care. The weight (ω_r) is calculated in a similar manner but r goes from -7 to 7.

The average expected number of infants in the hospital on the delivery day is 22 and the average expected number of infants during the hospital stay is 34. Panel B of Table II summarizes the residuals from regressions of the expected numbers of infant on delivery date and the expected numbers of infants during stay with a complete set of hospital and day of the week fixed effects. These measures of crowdedness also display significant variability. The respective standard deviations of the residuals are 4.51 and 5.82.

Table VII reports the first stage regression results of equation (1) using three alternative measures of crowdedness. Column (ii) reports regression result using the number of infants born on a given day as reported in Table IV. Columns (iii) and (iv) shows the results where crowdedness is measured by expected number of infants on the delivery date and expected number of infants during the hospital stay respectively. Both additional measures of hospital crowdedness reveal a negative and statistically significant relationship between hospital crowdedness and infant hospital charges. A one standard deviation decrease in crowdedness after netting out hospital and day of the week fixed effects corresponds to an increase in spending of approximately \$90 for each of the alternate measures of crowdedness. The F-statistic is somewhat weaker for the measure of crowdedness that includes deliveries that occur after the infant's date of birth. This is likely because the majority of charges occur on the delivery date so it is to be expected that births that occur in the near future have less of an impact on spending than births that occur on the delivery date or before. Overall, the first stage results provide additional evidence that health care providers change the intensity of treatment based on the short-term fluctuations in crowdedness in the hospital.

Panel B of Table VII reports the second stage regression results using the alternative crowdedness measures. Each cell in Table VII corresponds to a separate regression. Columns (i)

and (ii) replicate the results in Table V. Columns (iii) and (iv) confirm that additional spending caused by lessening of hospital crowdedness does not improve infant mortality. Across specifications, the range of the estimates is fairly tight and, in most cases, the marginal effect varies little. Regardless of how crowdedness is measured there is an insignificant but positive relationship between additional spending and infant mortality rates.

c. Low-birth-weight Infants

We now focus on the subsample of infants with birth-weight less than 2500 grams. The average neonatal mortality rate in our sample is very low at 0.3 percent. This low mortality rate could mask the effects of higher spending. Mortality for the low-birth-weight subsample is around 10 times higher than all infants. Almond et al. (2010) find a positive return to additional treatments for very-low-birth-weight newborns.¹⁴ Also, Figure I documents that differences in hospital charge between infants born on *busy* days and *slow* days are especially pronounced among low-birth-weight infants. We thus examine the impact of crowdedness for the subsample of infants with a birth weight less than 2500g. Because low-birth-weight infants are more likely to be transferred to another hospital on their delivery day and because the crowdedness of the birth hospital should not have any impact on infants who immediately transferred, we exclude low-birth-weight infants who are transferred to other hospital on their birth date from this analysis.¹⁵

¹⁴ Barreca et al. (2011) criticized the findings of Almond et al. (2010) that the regression discontinuity design used is sensitive to the exclusion of observations due to lumping of the birth weights at 1500g (there is a concern that less healthy newborns may be disproportionately likely to have their birth weight rounded to 1500g). Almond et al. (2011) respond that the lumping at 1500g is not likely to cause bias because data show that infants with birth weight of 1500g are less likely to receive intensive care.

¹⁵ When we exclude infants who transferred on their day of birth from our main analysis, the findings are similar to the main findings.

Panel A in Table VIII reports larger first stage estimates than our main sample for all three measures of crowdedness. When crowdedness decreases by one standard deviation, hospital spending per low-birth-weight infant increases by between \$1,463 and \$2,124 depending on the measure of crowdedness used. The first stage F-statistics are sufficiently large for all measures of crowdedness, and unlike the full sample they are larger when the broader measure of hospital crowdedness is used. This is likely because low-birth weight infants require intensive treatments beyond their delivery dates, as such a crowded day on a day following their birth directly impacts their treatment intensity. The first stage results in Table VIII confirm that when the hospital is more crowded, even the most vulnerable infants receive less intensive treatment. Furthermore, the impact of crowdedness on health care spending is sizable for this population.

Column (i) in Panel B of Table VIII reports the OLS estimates, which suggest that additional hospital spending on low-birth weight infants leads to a statistically significant decrease in infant mortality. Columns (ii)- (iv) contain the corresponding 2SLS estimates. They suggest that even for this vulnerable population, the additional spending associated with slower days does not improve infant health as measured by neonatal and one-year mortality rates. Although mostly statistically insignificant, the estimated marginal effects are all positive.

d. Including Mothers Hospital Charge

The OSHPD records charges to the infant and charges to the mother as separate entries. Mothers hospital charges include some charges (such as caesarian sections) that are directly related to the labor and delivery. Thus infants may also receive health benefit from the spending assigned to

their mothers. As a robustness check, we add mother charge to infant charge to examine the impact of all the charges related to the delivery.

Panel A of Table IX presents the impact of hospital crowdedness on the mother and infant charge. When the number of births is used as the crowdedness measure, the first stage results imply that a one standard deviation increase in the number of births in a given hospital-day of the week is associated with a decrease in average total hospital charge of mother and infant by \$110. If one compares the results from Table IX to the results from Table VII it is apparent that spending on the birth mother is responsive to variation in hospital crowdedness but not as responsive as spending on the infant. When the broadest measure of crowdedness is used as instrument there is a potential weak instrument problem since births that occur before and after the delivery date do not impact mothers charges associated with delivery.

Panel B of Table IX reports the impact of mother and infant charges using the three measures of crowdedness as instruments for total spending. Although OLS results again report a negative correlation between hospital spending (as measured by the sum of all infant and mother charges) and infant mortality, columns (ii)-(iv) confirm the main result that additional spending on mother and infant does not reduce neonatal or one-year mortality rates.

e. The Impact of Length of Hospital Stay

Although we believe hospital charge is the best available comprehensive measure of hospital care, it is possible that the lower hospital charges associated with busy day infants are due to recording bias. Specifically, hospitals may fail to record and bill all the procedures performed because the hospital personals are swarmed on busy days.¹⁶ If the recording bias is strong

¹⁶ We find no evidence of hospitals' failure to record procedures. Procedure data is available for 27.81 percent of infants born on busiest quartile days vs. 27.09 percent of infants born on slowest quartile days. We

enough, infants born on slow days and busy days might receive the similar level of hospital care (although they report different levels) which would account for our null findings of hospital spending on infant health in the second stage. To investigate the possibility of recording bias, we consider length of hospital stay which is less likely to be subject to recording bias. As we did with charges, if an infant is transferred to another California hospital we add those days to the days spent at the hospital of birth. We examine if hospital crowdedness has any impact on the length of hospital stay and if longer hospital stays have beneficial impact on infant health.

Panel A in Table X presents first stage results of crowdedness on length of stay using the three different measures of crowdedness. Although the magnitude is small, the first stage results are negative and statistically significant for all measures of crowdedness. Each additional birth during the target infant's hospital stay reduces the average length of stay. This confirms our main finding that hospital treatment intensity responds to short-term fluctuations in hospital crowdedness. The first stage F-statistics is the highest when the expected number of infants during stay is used suggesting the length of hospital stay is the most responsive to the number of infants born before and after.

Panel B presents OLS and 2SLS estimates of length of hospitalization stay (measured in days) on infant mortality. The OLS results, located in column (i), again suggest that additional treatment intensity is associated with a small reduction in infant mortality. Columns (ii)-(iv) report the second stage regression results on the impact of hospital stay on neonatal and one-year mortality. They are also consistent with our main findings. The OLS estimates of the treatment

also identify four commonly performed procedures – Circumcision (ICD9:640), Vaccination NEC(ICD: 9955), Insertion of endotracheal tube (ICD9: 9604), and Other phototherapy (other than ultraviolet, ICD:9983) . The difference in percentage of infant receiving each procedure across *slow* and *busy* days is smaller than 1 percentage point for all four procedures. Moreover, except Circumcision, which is more physician-intensive, the percentages of newborns reporting the other three most common procedures are higher among infants born on busier quartile days. This suggests that hospitals are consistently recording the procedures regardless of hospital crowdedness.

on mortality are biased downward. Shorter stays that arise because of hospital crowdedness do not adversely impact either neonatal or one-year mortality.

VI. Discussion

It is possible that the null health effects documented above are simply an artifact of supplier-induced demand. This change in treatment decisions in the presence of income threat is well documented in the supplier induced demand literature.¹⁷ If supplier-induced demand is driving the additional spending that occurs on slow days, then one would expect little to no benefit from the additional treatments since the additional charges are only occurring to pad the medical bills and not because they are deemed medically necessary.¹⁸ In addition to the finding documented earlier that the response of spending to crowdedness is greater in more capacity constrained hospitals, we conduct the following two exercises which suggest that the results do not simply reflect supplier-induced demand and instead are consistent with the relaxation of binding capacity constraints.

First, we split the sample into deliveries that were covered by a managed care plan and those which were not. If supplier-induced demand is driving the negative relationship between crowdedness and spending, then we should expect a smaller or non-existent first-stage for those births covered by managed care. As is shown in Table XI there is no difference in the response

¹⁷ One of the features of medical market is an agency relationship between doctors and patients. Because doctors have asymmetrically more knowledge about medical care than their patients do, doctors are expected to behave as patients' agents when making treatment decisions. However, studies find that when doctors face negative income shocks, doctors may exploit the agency relationship and provide more care in order to maintain their income. See Chapter 9 - Physician Agency from the Handbook of Health Economics (McGuire, 2000) and Gruber and Owings (1996) for reviews of the supplier induced demand literature.

¹⁸ An additional explanation for the null findings in the second stage is that first stage results are being driven by teaching hospitals allowing residents perform extra procedures on slow days. We split the sample into teaching and non-teaching hospitals and find that if anything, the first stage is stronger for non-teaching hospitals.

of hospital charges to crowdedness on the day of birth by managed care status.¹⁹ While both samples have a strong first stage, in neither sample do we observe positive health benefits from additional charges, although the marginal benefit of additional spending may be larger for managed care births.

Second, under the capacity constraint hypothesis, at low levels of crowdedness additional births should not impact the amount of care (since the hospital still has ample capacity). At some point the hospital will exceed its normal capacity after which additional births will reduce spending per birth. Under a supplier-induced demand model, on the slowest days the hospital personnel face the largest reduction in income, as such spending should be greater at very low levels of crowdedness than at low-to-moderate levels of crowdedness.

To investigate this, we estimate our first stage model equation (1) but replace the daily crowdedness measure with nine crowdedness deciles constructed within hospital, day of the week, and year. So the 90th decile would turn on if an infant was born on one of the five busiest Mondays (or any other day of the week) in a given hospital in a given year. Figure III shows the results of this exercise graphically where the omitted decile is the slowest decile. As shown in Figure III, infants born on days that are below the median level of crowdedness for a given day of the week, hospital, and year experience the same treatment intensity. However, once the hospital exceeds the median level of crowdedness then as the hospital becomes more crowded spending declines. These findings are inconsistent with most supplier-induced demand models and suggest instead that the hospitals face binding capacity constraints. In other words, the care that is cut is care that the hospital provides whenever it has the capacity to do so.

¹⁹ In results not shown that use the alternative measures of crowdedness, the range of the estimates is fairly tight and the marginal effect varies little by insurance coverage. We also split the sample by not-for-profit status with the hypothesis that supplier induced demand might be larger at for-profit hospitals. We find no difference in the response of hospital charges to crowdedness by not-for-profit status.

VII. Conclusions

Using the hospital level of crowdedness on an infant's birth date to generate exogenous variation in treatment intensity, we estimate the impact of additional health care measured by hospital spending on infants, delivery spending, and length of hospital stay on infant health for the universe of birth in California hospitals between the years 2002 and 2006. Our paper documents that on days when more deliveries occur than average, hospitals respond to the short-term crowdedness by treating infant less intensively. We present evidence that supports the use of crowdedness as an instrument for treatment intensity. First, infants born on busy days receive significantly less care than infants born on slow days with identical underlying health status. Second, we show that the impact of crowdedness on treatment intensity is larger in more resource constrained hospitals. Third, we show that crowdedness is unrelated to mother or infant characteristics which may impact infant health.

We find that the additional hospital spending on infants born on slower days does not translate to better infant health outcomes measured by neonatal and one-year mortality rates. This is in contrast to the OLS estimates which consistently suggest that additional treatments reduce mortality. These findings are robust to alternative measures of hospital crowdedness and alternative measures of hospital treatment.

Our results identify the health consequences of hospital care on infants who received additional treatments solely because they were born on slow days when the hospital had excess capacity. If medical personnel allocate care on the basis of unobservable sickness, then the infants who received additional care when capacity constraints are relaxed will be unobservably healthier. In this case, estimates of the relationship between treatment and health would be biased towards finding a positive effect of treatment. Given that we find no health benefits from

additional treatment, we can conclude that on the margin, additional spending does not reduce infant mortality. Our study suggests that we are at the so-called “flat part of the curve”.

Unlike other studies that utilize variation in treatment intensity from legislative changes, the variation in treatment identified in our study stems from the medical personnel’s decisions. We identify health consequences of reduction in care for infants born on busier days in the same hospital with identical health and family background characteristics. This is precisely the information from which the policy debates about staffing ratio and the number of newborn intensive care units benefit. Our findings suggest that new hospital construction, changes in staffing requirements, or reduction in birth rates that reduce crowdedness in maternity wards will lead to additional per infant hospital charges without increasing infant health status.

Our result should be interpreted cautiously. While we measure infant health outcomes using neonatal and one-year mortality rates, and these are the best available proxy for infant health, they are not the perfect measures of health status. It is possible that additional spending has benefits such as decreased levels of discomfort for infants or improved parental satisfaction that we are unable to measure. Additionally, because resources and commitment to prenatal health may differ across states and countries, our results might not be generalized beyond California. In particular, our results may not generalize to states that have certificate of need laws, which require state approval for hospital construction or expansion, in place.

Finally we note that the instrument which we propose in this paper, hospital crowdedness, could be applied other settings where there is little possibility of timing the onset of medical need. For example, the impact of emergency room care for heart attack patients could benefit from employing a hospital crowdedness measure as an exogenous source of variation to access to treatment.

Author affiliation:

Marks: University of California, Riverside

Choi: RAND/University of California, Los Angeles

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Table I. Summary Statistics

Variable	<u>All Births</u>		<u>Low Birth Weight (<2500g)</u>	
	Mean	S.D.	Mean	S.D.
Pregnancy Characteristics				
Month prenatal care began	2.24	(1.40)	2.18	(1.41)
Number of prenatal visits	12.40	(4.03)	11.90	(5.69)
No pregnancy complication	0.34	(0.47)	0.28	(0.45)
Parents' Characteristics				
Mother's age	27.92	(6.37)	28.41	(6.94)
Mother's education (years)	12.23	(3.42)	12.33	(3.33)
Father's age	30.85	(7.20)	31.25	(7.78)
Father's education (years)	12.19	(3.59)	12.27	(3.55)
Newborn Characteristics				
Boy	0.51	(0.50)	0.48	(0.50)
White	0.71	(0.45)	0.64	(0.48)
Black	0.07	(0.25)	0.12	(0.32)
Asian	0.08	(0.28)	0.09	(0.29)
Hispanic	0.52	(0.50)	0.47	(0.50)
First born	0.39	(0.49)	0.41	(0.49)
Single birth	0.97	(0.17)	0.75	(0.43)
Birth Characteristics				
Birth weight (g)	3,320	(572)	1,986	(515)
Gestation (days)	275	(24)	247	(38)
Caesarean section	0.31	(0.46)	0.53	(0.50)
Primary Payer				
Medicaid	0.51	(0.50)	0.49	(0.50)
Private insurance	0.44	(0.50)	0.43	(0.50)
Self-pay	0.02	(0.15)	0.02	(0.15)
Variables of Interest				
Infant hospital charges (\$)	12,985	(78,210)	114,034	(256,057)
Mother and infant hospital charges	27,041	(82,338)	139,721	(265,577)
Hospital stay (days)	3.36	(8.04)	16.50	(25.36)
Number of procedures ¹	0.43	(1.06)	1.68	(2.81)
Probability of transfer from birth hospital	0.017	(0.130)	0.107	(0.309)
Probability of the same day transfer	0.009	(0.094)	0.063	(0.243)
Outcome Variables				
Neonatal mortality	0.003	(0.057)	0.038	(0.191)
One-year mortality	0.005	(0.069)	0.047	(0.211)
Observations	2,287,119		150,801	

1. Data on procedures done are available for only 28 percent of the observations.

Table II. Descriptive Statistics of Number of Births for Selected Hospitals

	Mean (Daily Births)	Standard Deviation	Min	25 th Percentile	50 th Percentile	75 th Percentile	Max
<u>Panel A</u>							
Smallest Hospital	2	0.98	1	1	2	3	5
10 Percentile Hospital	4.35	1.86	1	3	4	6	10
20 Percentile Hospital	5.79	2.31	1	4	6	7	13
30 Percentile Hospital	6.94	2.58	1	5	7	9	16
40 Percentile Hospital	8.01	2.89	1	6	8	10	17
50 Percentile Hospital	8.76	3.25	1	6	9	11	20
60 Percentile Hospital	9.94	2.99	1	8	10	12	20
70 Percentile Hospital	11.57	3.66	1	9	12	13	25
80 Percentile Hospital	13.80	3.67	3	11	14	16	26
90 Percentile Hospital	17.43	4.40	3	14	17	20	31
Biggest Hospital	22.37	5.79	4	15	23	26	42
<u>Panel B</u>							
Residual Crowdedness:							
Number of Births	0	3.13	-17.73	-1.97	-0.13	1.84	20.09
Expected Num. of Infants on Birth Date	0	4.51	-39.77	-2.69	-0.11	2.60	28.89
Expected Num. of Infants during Hospital Stay	0	5.82	-60.44	-3.34	-0.12	3.28	37.31
Number of Hospitals	226						

Note: Residual crowdedness is the residuals from separate regressions of hospital crowdedness on the complete set of hospital and day of the week fixed effects for each of the three crowdedness measure.

Table III. Summary Statistics of Infants Born on Top 25% Slowest and Busiest Days

Variable	<u>Slow Days</u>		<u>Busy Days</u>		Difference
	Mean	S.D.	Mean	S.D.	
Number of infants born	7.45	(4.53)	15.05	(6.40)	-7.60
Same day transfer	0.008	(0.090)	0.009	(0.093)	-0.001
Same day transfer: low birth weight	0.06	(0.24)	0.06	(0.24)	0.001
Pregnancy Characteristics					
Trimester prenatal care began	2.21	(1.38)	2.21	(1.38)	0.00
Number of prenatal visits	12.47	(3.96)	12.53	(3.96)	-0.06
No pregnancy complication	0.35	(0.48)	0.36	(0.48)	-0.01
Parents' Characteristics					
Mother's age	27.95	(6.39)	27.98	(6.39)	-0.03
Mother's education (years)	12.22	(3.42)	12.27	(3.42)	-0.05
Father's age	30.89	(7.21)	30.91	(7.21)	-0.02
Father's education (years)	12.18	(3.59)	12.23	(3.59)	-0.05
Newborn Characteristics					
Boy	0.51	(0.50)	0.51	(0.50)	0.00
White	0.71	(0.46)	0.71	(0.46)	0.00
Black	0.07	(0.26)	0.07	(0.26)	0.00
Asian	0.09	(0.29)	0.09	(0.29)	0.00
Hispanic	0.52	(0.50)	0.53	(0.50)	-0.01
First born	0.39	(0.49)	0.38	(0.49)	0.00
Multiple birth (twin or more)	0.02	(0.14)	0.04	(0.14)	-0.02
Primary Payer					
Medicaid	0.51	(0.50)	0.52	(0.50)	0.00
Private insurance	0.45	(0.50)	0.44	(0.50)	0.00
Self-pay	0.02	(0.15)	0.02	(0.15)	0.00
Birth Characteristics					
Birth weight (g)	3,328	(570)	3,296	(569)	31.72
Gestation (days)	275	(24)	275	(24)	0
Caesarean section	0.31	(0.46)	0.32	(0.46)	-0.01
Observations	571,615		571,790		

Note: *Slow /busy* days are days with residual crowdedness in the bottom/top quartile of the residuals from a regression of number of daily births on the complete set of hospital and day of the week fixed effects.

Table IV. Impact of Hospital Crowdedness on Hospital Spending

	Number of Births
Coefficient	-0.0016
Robust Standard Error	(0.0004)**
F-Statistics	19.60
R-squared	0.5214
	[\$ 65.03]
Observations	2,287,119

Notes: Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. Regression includes the following control variables: mother and father's age and education and indicators for day of the week, month, year, holiday, birth weight categorized 500g interval (less than 500g, 501-1000g,..., 4001-4500, over 4500g), gestation in two weeks interval (less than 32 weeks, 33-34 weeks,..., over 43 weeks), infant's race, gender, parity, month prenatal care began, number of prenatal visits, caesarean section, and insurance status. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness). *denotes significant at 5%; ** significant at 1%.

Table V. Impact of Hospital Spending on Infant Health

	OLS Results	2SLS Results
	(i)	(ii)
<u>Neonatal Mortality</u>		
Log(Health Care Spending)	-0.0003 (0.0000)**	0.0022 (0.0024)
<u>One-year Mortality</u>		
Log(Health Care Spending)	-0.0001 (0.0000)*	0.0031 (0.0050)
Observations	2,287,119	2,287,119

Notes: Health Care Spending is the sum of infant hospital charges from the birth hospital and any consecutive hospital stay during the first hospitalization. Each cell represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in the table are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses.

Table VI. Capacity Heterogeneity

	Low-capacity Hospitals (i)	High-capacity Hospitals (ii)
<u>Panel A: 1st Stage Results</u>		
Coefficient	-0.0025	-0.0008
Robust S.E.	(0.0004)**	(0.0005)
F-Statistics	34.78	2.62
R-squared	0.4839	0.5346
	[\$ 66.62]	[\$ 47.50]
<u>Panel B: 2nd Stage Results</u>		
	<u>Neonatal Mortality</u>	
Log(Health Care Spending)	-0.0017 (0.0026)	0.0109 (0.0060)
	<u>One-year Mortality</u>	
Log(Health Care Spending)	-0.0032 (0.0054)	0.0156 (0.0122)
<u>Panel C: Summary Statistics</u>		
Ave. Number of Births	8.67	11.19
Ave. Birth Weight	3343	3296
Ave. Gestation	276	275
Ave. Infant Charges	\$ 8,051	\$ 17,939
Ave. Mother and Infant Charges	\$ 20,841	\$ 33,267
Ave. Hospital Stay (Days)	2.81	3.91
NICU Beds per Thousand	1.46	9.19
Number of Hospitals	152	98
Observations	1,145,902	1,141,217

Notes: Twenty four hospitals belong to both high and low capacity depending on the years. One hundred eight hospitals have no NICU beds and are included in the low capacity category. The number of NICU beds data is obtained from Hospital Annual Utilization DATA available at http://www.oshpd.ca.gov/hid/Products/Hospitals/Utilization/Hospital_Utilization.html. Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in Panel B are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. Standard errors are in parentheses and clustered at the hospital X year level. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Table VII. Impact of Alternative Measures of Crowdedness

	OLS Results	2SLS Results		
	(i)	Number of Births (ii)	Expected Number of Newborns on Delivery Day (iii)	Expected Number of Newborns during Stay (iv)
<u>Panel A: 1st Stage Results</u>				
Coefficient		-0.0016	-0.0016	-0.0012
Robust S.E.		(0.0004)**	(0.0004)**	(0.0004)**
F-Statistics		19.60	17.46	10.71
R-squared		0.5214	0.5214	0.5214
		[\$ 65.03]	[\$ 93.70]	[\$ 90.69]
<u>Panel B: 2nd Stage Results</u>				
			<u>Neonatal Mortality</u>	
Log(Health Care Spending)	-0.0003 (0.0000)**	0.0022 (0.0024)	0.0027 (0.0017)	0.0010 (0.0017)
			<u>One-year Mortality</u>	
Log(Health Care Spending)	-0.0001 (0.0000)*	0.0031 (0.0050)	0.0063 (0.0032)	0.0050 (0.0033)
Observations	2,287,119	2,287,119	2,287,119	2,287,119

Notes: Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in the Panel B are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Table VIII. Impact of Hospital Spending on Low-Birth-Weight Infants (<2500g)

	OLS Results	2SLS Results		
	(i)	Number of Births (ii)	Expected Number of Newborns on Delivery Day (iii)	Expected Number of Newborns during Stay (iv)
<u>Panel A: 1st Stage Results</u>				
Coefficient		-0.0041	-0.0040	-0.0032
Robust Standard Error		(0.0012)**	(0.0008)**	(0.0007)**
F-Statistics		11.12	24.09	23.82
R-squared		0.6402	0.6403	0.6403
		[\$ 1,463.40]	[\$ 2,057.18]	[\$ 2,123.78]
<u>Panel B: 2nd Stage Results</u>				
			<u>Neonatal Mortality</u>	
Log(Health Care Spending)	-0.0037 (0.0001)**	0.01400 (0.0131)	0.0197 (0.0092)*	0.0078 (0.0092)
			<u>One-year Mortality</u>	
Log(Health Care Spending)	-0.0029 (0.0003)**	0.0056 (0.0189)	0.0183 (0.0125)	0.0071 (0.0121)
Observations	137,290	137,290	137,290	137,290

Notes: Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in the Panel B are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Table IX. Impact of Mother + Infant Hospital Charges

	OLS Results	2SLS Results		
		Number of Births	Expected Number of Newborns on Delivery Day	Expected Number of Newborns during Stay
	(i)	(ii)	(iii)	(iv)
<u>Panel A: 1st Stage Results</u>				
Coefficient		-0.0013	-0.0011	-0.0009
Robust Standard Error		(0.0003)**	(0.0003)**	(0.0003)**
F-Statistics		20.76	11.42	7.31
R-squared		0.6497	0.6497	0.6497
		[\$ 110.03]	[\$ 134.15]	[\$ 141.64]
<u>Panel B: 2nd Stage Results</u>				
			<u>Neonatal Mortality</u>	
Log(Infant + Mother Charges)	-0.0007 (0.0000)**	0.0026 (0.0029)	0.0038 (0.0023)	0.0014 (0.0023)
			<u>One-year Mortality</u>	
Log(Infant + Mother Charges)	-0.0001 (0.0001)*	0.0036 (0.0059)	0.0088 (0.0045)	0.0068 (0.0045)
Observations	2,287,119	2,287,119	2,287,119	2,287,119

Notes: Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in the Panel B are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Table X. Impact of Hospital Stay

	OLS Results	2SLS Results		
	(i)	Number of Births (ii)	Expected Number of Newborns on Delivery Day (iii)	Expected Number of Newborns during Stay (iv)
<u>Panel A: 1st Stage Results</u>				
Coefficient		-0.0058	-0.0061	-0.0053
Robust S.E.		(0.0014)**	(0.0011)**	(0.0009)**
F-Statistics		16.15	29.25	32.38
R-squared		0.4728	0.4728	0.4728
		[-0.06 days]	[-0.09 days]	[-0.10 days]
<u>Panel B: 2nd Stage Results</u>				
			<u>Neonatal Mortality</u>	
Hospital Stay (1Day)	-0.0001 (0.0000)**	0.0006 (0.0007)	0.0007 (0.0004)	0.0002 (0.0004)
			<u>One-year Mortality</u>	
Hospital Stay (1Day)	-0.0001 (0.0000)**	0.0008 (0.0014)	0.0016 (0.0008)	0.0011 (0.0007)
Observations	2,287,119	2,287,119	2,287,119	2,287,119

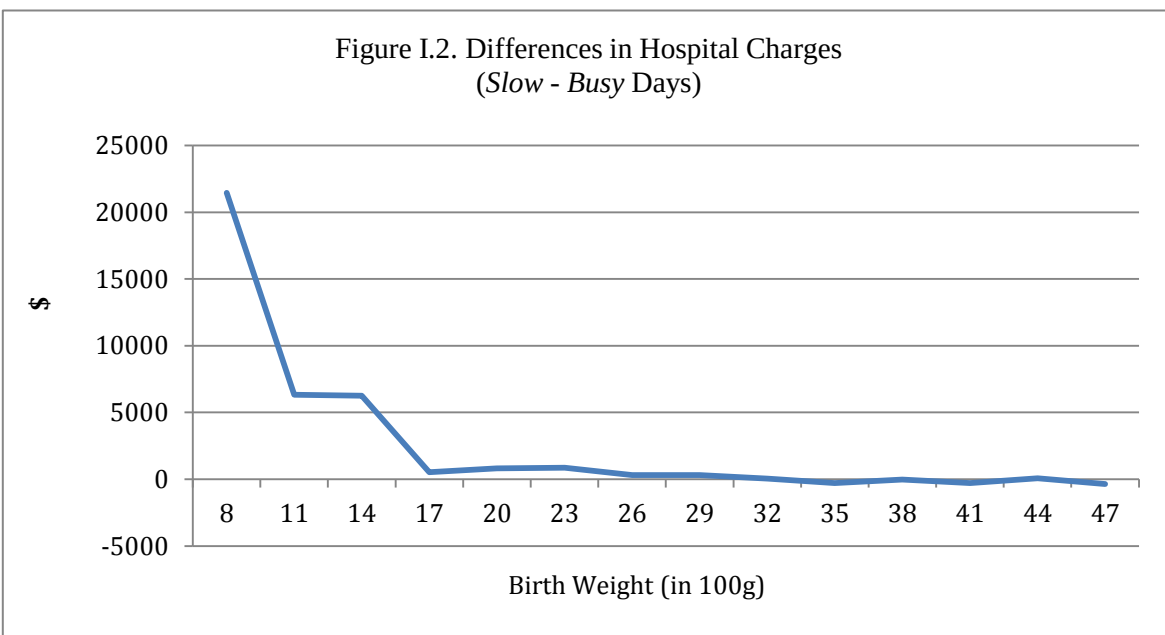
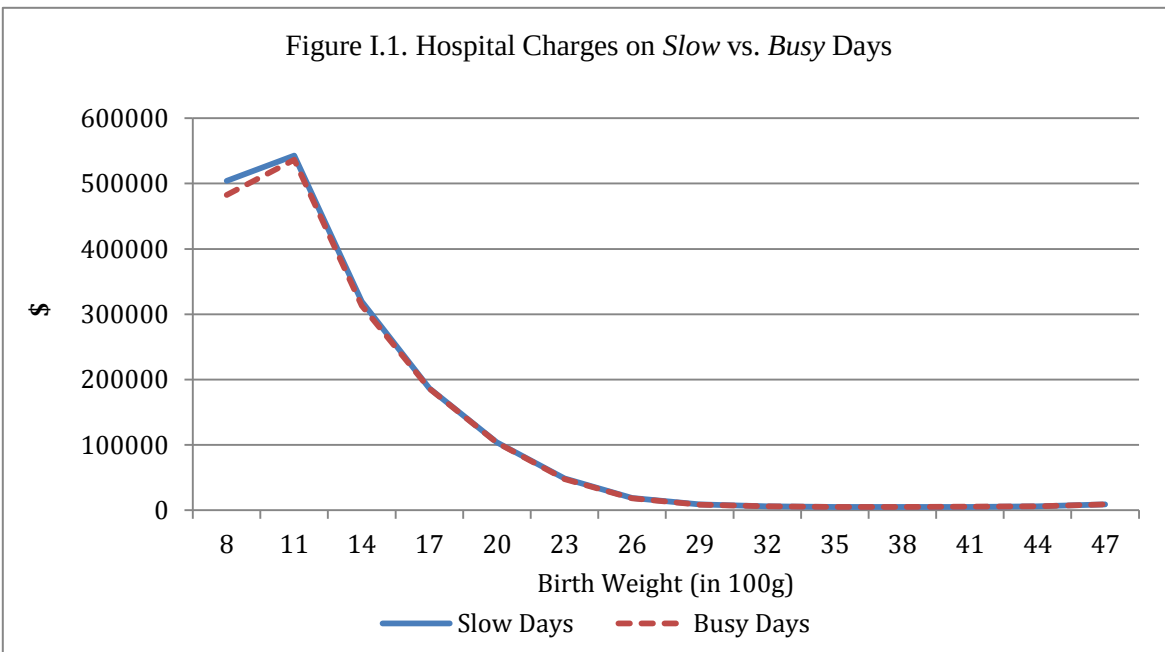
Notes: The dependent variable is the length of the infant's first hospital stay including transfers. Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV. The marginal effects presented in the Panel B are evaluated at the mean. *denotes significant at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. The number in bracket corresponds to the predicted change in average hospital stays from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Table XI. Managed Care vs. Traditional Insurance

	Managed Care (i)	Traditional Insurance (ii)
<u>Panel A: 1st Stage Results</u>		
Coefficient	-0.0017	-0.0014
Robust Standard Error	(0.0004)**	(0.0004)**
F-Statistics	18.80	12.64
R-squared	0.5218	0.521
	[\$ 68.99]	[\$ 57.02]
<u>Panel B: 2nd Stage Results</u>		
	<u>Neonatal Mortality</u>	
Log(Health Care Spending)	-0.0015 (0.0030)	0.0081 (0.0027)*
	<u>One-year Mortality</u>	
Log(Health Care Spending)	-0.0016 (0.0059)	0.0109 (0.0076)
<u>Panel C: Summary Statistics</u>		
Ave. Number of Births	9.93	9.93
Ave. Birth Weight	3,320	3,319
Ave. Gestation	275	275
Ave. Infant Charges	\$ 12,965	\$13,013
Ave. Mother and Infant Charges	\$ 27,006	\$ 27,012
Ave. Hospital Stay (Days)	3.36	3.36
Observations	1,353,713	933,406

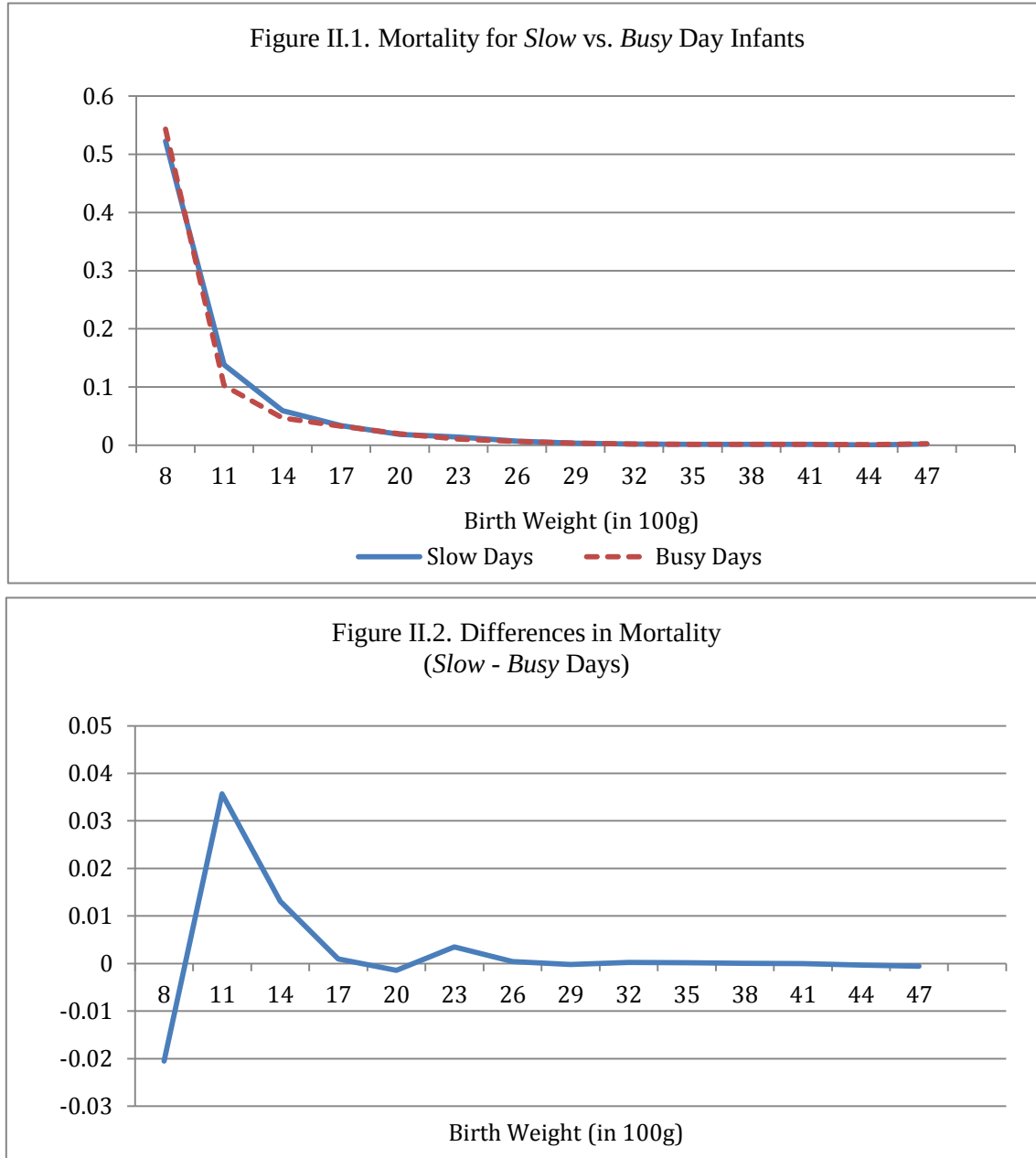
Notes: Each cell in panel B represents a separate regression. All models include the control variables listed in Table IV except insurance status. The marginal effects presented in the Panel B are evaluated at the mean. *denotes significant at at 5%; ** significant at 1%. Robust standard errors corrected for clustering on hospital and year cells are reported in parentheses. The number in bracket corresponds to the predicted change in average hospital charges from a one standard deviation change in the number of births in a given hospital-day of the week (residual crowdedness).

Figure I. Hospital Charges on Infants Born on *Slow* vs. *Busy* Days



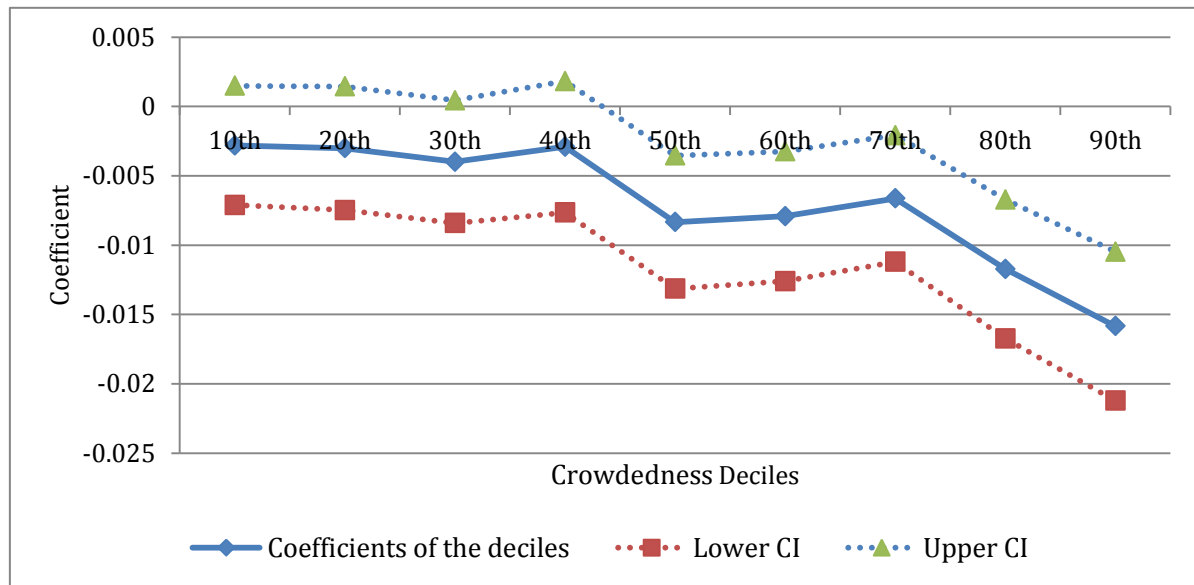
Notes: *Slow* /*busy* days are days with residual crowdedness in the bottom/top quartile of the residuals from a regression of number of daily births on the complete set of hospital and day of the week fixed effects. Each point corresponds to the average hospital charges for all slow or busy day births within a 300 gram birth weight bin starting at 800 grams. Less than 800g and more than 4700g infants are grouped together.

Figure II. One-year Mortality for Infants Born on *Slow* vs. *Busy* Days



Note: See Figure I for variable description.

Figure III. Hospital Spending by Crowdedness Decile



Notes: To construct the crowdedness deciles, we first divided the sample into 7735 hospitals, days of the week, and year cells. Fourteen hospitals shut down during the sample years, therefore they are only included in some years. Within each cell, we ordered the daily number of infants born and constructed crowdedness deciles. Each point is the coefficient estimate of the corresponding decile crowdedness indicator in a regression with log hospital charges as the independent variable and full set of control variables listed in Table IV. The coefficients are relative to the bottom decile (the least crowded days), which is omitted. The dotted lines are 95 percentile confidence intervals.